

Proof by contradiction

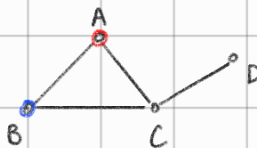
- 1) suppose false
- 2) manipulate until you reach contradiction
 - ground truths
 - assumption claim

ex. Prove by contradiction that there is no largest even integer

Proof: Suppose there is a largest even integer x . By definition of even integers, $x = 2y$, $y \in \mathbb{Z}$. Let's consider $z = 2y + 2$, we know that $z > x$ because $2y + 2 > 2y$. $z = 2(y + 1)$, so by definition of even, z is also even. Thus z is an even integer larger than x , and we have contradiction.

Lower Bound of Chromatic Number

• Example of contradiction

ex.  Prove by contradiction that chromatic number is at least greater than 2

Proof: Suppose we can color this graph with two colors. Suppose A is red and B is blue. Then C cannot be red or blue by definition of graph color. Therefore, we need a third color for C. Then we have a contradiction, so it is impossible to color this graph with only two colors and the chromatic number is ≥ 3 . $\square$

Irrational Numbers

• Another common application of proof by contradiction

ex. We know $\sqrt{6}$ is irrational, prove $\sqrt{2} + \sqrt{3}$ is irrational

Proof: Suppose $\sqrt{2} + \sqrt{3}$ is rational. Then $\exists p, q \in \mathbb{Z}, q \neq 0$, st $\sqrt{2} + \sqrt{3} = \frac{p}{q}$.
 $(\sqrt{2} + \sqrt{3})^2 = \frac{p^2}{q^2}$, $2 + 3 + 2\sqrt{6} = \frac{p^2}{q^2}$, $2\sqrt{6} = \frac{p^2}{q^2} - 5$, $\sqrt{6} = \frac{p^2 - 5q^2}{2q^2}$. Since $p, q \in \mathbb{Z}$, $p^2 - 5q^2 \in \mathbb{Z}$ and $2q^2 \in \mathbb{Z}$. Then, $\sqrt{6}$ would be rational, which is false and is a contradiction. Thus, $\sqrt{2} + \sqrt{3}$ must be irrational.

Proof by Contradiction vs Proof by Contrapositive

Contradiction

- Suppose given claim p
- supposing $\neg p$
- don't need x implies

y form

- existential proofs
- rational/irrational numbers

Contrapositive

- $p \rightarrow q$
- show $\neg q \rightarrow \neg p$